ARGUS Project

Mission Definition, Payload Description

ARGUS project – November 2026

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1 Mission Definition

1.1 Studied object / functional mission target

Definitive choice: space debris in low Earth orbit, a small object (about 1 kg, aluminium surface) to be tracked and deflected by a CubeSat. **[TO DO: confirm the precise object or the precise part of the object (e.g. type of debris, size, rotation)]**

1.2 Scientific / technological question or aim

Definitive choice: demonstrate that a 6U CubeSat can detect and track a small debris at short range with a LiDAR, and change its trajectory by laser ablation (target small ΔV,F $= 30\,mNon1\,kg$).

1.3 Main payload

Definitive choice: pulsed ablation LASER (1064 nm, class 4) guided by a LiDAR (1550 nm, SPAD detector, MEMS micro-mirror). **[TO DO: confirm the final laser parameters: E_p, P_c, repetition rate, average power]**

1.4 Secondary payloads (if any)

[TO DO: list the secondary payloads, ML and camera]

1.5 Experts contacted

Ratio of contacted experts: **[TO DO: X]** % (**[TO DO: X]** experts contacted out of **[TO DO: Y]** experts initially identified).
[TO DO: table of experts (name, field, topic, contacted or not) – to be defined once the other subsystem is discussed] Vincent robert for space mecanique if u have someone for laser ablation and ml and camera, please contact him

2 Payload Description

2.1 Functional description of the payloads' behavior

[TO DO: complete and check the sequence below]

1. **Detection:** the LiDAR scans its field of view ($\pm 10°$) and detects the debris.
2. **Tracking:** the LiDAR measures the position of the debris and keeps tracking it; the fine pointing mirror is guided by the LiDAR data.
3. **Lock check:** firing is authorized only if the target is locked (shutter opened).
4. **Firing:** short laser bursts on battery power, ablating the aluminium surface to produce thrust.
5. **Monitoring:** the debris is tracked during and after each burst; firing stops automatically in case of anomaly (fluence, loss of tracking).
6. **Safe state:** firing inhibited by default outside the contact windows or without lock.

2.2 Quantified description of the payloads' performance

[TO DO: quantified performance table of all payloads – to be discussed with the other subsystem]

3 System Architecture and Diagrams

3.1 Data Management Plan – functional diagram V1

Diagram made with JIRA (see the "Examples" tab). **[TO DO: insert the figure or the Jira link]**

3.2 Satellite platform – functional diagram V1

[TO DO: insert the figure (see the "Examples" tab)look on miro]

3.3 Satellite system model V1 (IDM-CIC)

[TO DO: insert the screenshot or the link to the IDM-CIC model]

4 Mission Analysis V1

[TO DO: orbit and mission phases – to be discussed with the other subsystem] [TO DO: maybe add some seach on procurreezy and vector ifyou have sometjing interresing]

5 Mission Report Summary V1

[TO DO: write the summary following the "Leads for Report Summary" tab: context, objectives, payloads, architecture, orbit and phases, main risks, schedule]

6 Risk Analysis

6.1 Risks related to the power LASER

Risk	Cause	Consequence	Level	Prevention
Power and heat too high for a 6U	1 kW optical with 35 % efficiency requires about 2.9 kW electrical, including nearly 1.9 kW of heat	Difficult to power and cool the system	High	Reduce P_{avg}, favour short shots on battery, perform a thermal study
Poor pointing	3 mm spot at 50 m, i.e. about 60 μrad, very demanding for a CubeSat	Shot misses the debris, no ablation	High	Fine pointing mirror guided by the LiDAR
Debris drifts away during firing	With $F = 30$ mN on 1 kg, the debris moves away by about 150 m in 100 s	Impossible to fire 2433 s continuously to reach 73 m/s	High	Short shots, debris tracking, aim for a smaller ΔV
Ablation efficiency overestimated	$C_m = 30$ μN·s/J assumed constant	Lower force and longer firing time	Medium	Take a margin, perform ground tests
Damaged optics	100 MW peak pulses, aluminium vapour deposition	Power loss, damaged optics	Medium	Margin on damage threshold, protective window
Debris fragmentation	Fluence too high or tumbling debris	New debris (Kessler syndrome)	High	Control the fluence, stop firing in case of anomaly
Unintended shot or reflection	Highly reflective aluminium surface	Dazzling or damaging another satellite	Medium	Shutter, firing authorized only if the target is locked
Hazard to people during tests	Class 4 laser at 1064 nm	Eye or skin injury	High	Suitable goggles, closed room, laser safety officer
Source failure	Ageing of the pump diodes	Loss of the mission	Medium	Operating margin, power monitoring

6.2 Risks related to the LiDAR

Risk	Cause	Consequence	Level	Prevention
Millimetre accuracy not reached	3 ns pulse and 200 ps SPAD jitter, i.e. about 3 cm	Position less accurate than expected	Medium	Announce centimetre accuracy, justify millimetre level by averaging
Return signal too weak	Small debris, SPAD efficiency of only 20 %, aluminium surface	Debris not detected	Medium	Perform a link budget, adjust energy or frequency
Sensor dazzling	Ablation laser, plasma, Sun or Earth in the field of view	Blind detection	Medium	1550 nm filter, gated operation, shutter

Risk	Cause	Consequence	Level	Prevention
Loss of debris tracking	Small tumbling target, field limited to ±10°	Firing impossible, target lost	High	Reacquisition algorithm, adapted field of view
Sensor degraded by radiation	Increasing dark noise (10 kHz at start)	False detections	Medium	Cooling, shielding, noise monitoring
Fragile MEMS micro-mirror	Vibration effects at launch	Mirror stuck or shifted	Medium	Launch lock, vibration and thermal tests
Synchronization and onboard computing	Offset between emission and detection, limited CubeSat computer	Wrong or too slow measurements	Medium	Common clock, lightweight algorithms
LiDAR eye safety	10 μJ beam at 20 kHz	Low hazard during tests	Low	Check the standard, limit access

6.3 Risks related to the camera and Machine Learning

Risk	Cause	Consequence	Level	Prevention
Missed or false debris detection	Small target, stars or reflections mistaken for debris	Target not acquired or wrong object tracked	High	Train with and without debris, check detection over several images
Poor image quality	Shadows, strong reflections or motion blur	Debris features cannot be located reliably	Medium	Adjust exposure, test different lighting conditions, reject unusable images
Training images not representative	Mostly simulated images, limited real views of the debris	Good ground results but poor performance in flight	High	Add real images, vary lighting and rotation, keep a separate test set
Debris too small in the image	Camera sized for 2–10 m, target farther away during firing	Detection possible but orientation cannot be estimated	High	Agree on a common operating distance, check target size in pixels
Wrong distance or orientation estimate	Symmetric debris, poorly located features or camera calibration error	Incorrect tracking and pointing data	High	Calibrate the camera, check the 3D model, compare distance with the LiDAR
Loss of visual tracking	Tumbling debris or target leaving the predicted image region	Position estimate no longer valid, firing interrupted	High	Restart full-image detection, inhibit firing until tracking is confirmed
Onboard processing too slow	Two neural networks and tracking running on limited hardware	Two observations per second not reached, delayed pointing data	Medium	Test the complete chain on Q8S and XE2, simplify the models if needed
Model deployment failure	Network operations unsupported on XE2 or reduced numerical precision	Model fails to run or gives less accurate results	Medium	Compile both networks early, compare onboard results with ground results

Risk	Cause	Consequence	Level	Prevention
Camera and LiDAR data out of sync	Different timestamps, transfer delays or inconsistent reference frames	Measurements refer to different target positions	High	Use a common clock, define reference frames, test the complete sensor interface

6.4 Risks related to the whole LiDAR + LASER payload

Risk	Cause	Consequence	Level	Prevention
Payload too heavy or power-hungry for a 6U	Laser, cooling, battery, LiDAR (about 12 W) and optics to accommodate	Mission not feasible in 6U	High	Mass, volume and power budget with the other teams
LiDAR / laser misalignment	Launch vibrations, temperature variations	Laser does not aim where the LiDAR measures	High	Common rigid structure, in-orbit calibration
Firing software bug	Error in the logic between detection and firing	Unintended shot or lock-up	High	Simple logic, loop testing
Electromagnetic interference	Pulsed laser power supply close to the sensor and the onboard computer	Disturbed measurements, reboots	Medium	Shielding, careful cabling, EMC tests
Shocks, vibrations and vacuum	Launch and orbital conditions	Misaligned or broken parts	Medium	Vibration, thermal and vacuum tests
Launcher safety requirements	Class 4 laser and battery on board	Integration refused	High	Laser disabled until separation, early dialogue with the launcher
Poor debris ejection	Velocity of 0.1 to 0.5 m/s poorly controlled	Debris returning towards the CubeSat	Medium	Simulate the relative trajectory before flight
Loss of communication with the ground	Firing outside the contact windows	Uncontrolled firing	Medium	Firing inhibited by default, safe autonomous sequence
Laser cost	Space-qualified 1 kW pulsed source beyond a student budget	Project blocked	High	Get quotes from suppliers, plan a lower-power demonstrator

6.5 Human and organizational risks (related to the subsystem)

Risk	Cause	Consequence	Level	Prevention
Unnoticed calculation errors	Different values in different places of the document	Wrong results in the report	Medium	Single calculation file, review by a colleague

Risk	Cause	Consequence	Level	Prevention
Limited skills	Little experience in laser, optics and laser safety	Poor technical choices, risk during tests	High	Training, advice from the supervisor or an expert
Poor communication between teams	Interfaces with power, thermal, structure and attitude not defined	Incompatibilities discovered too late	Medium	Regular meetings, interface document
Optimism on assumptions	Values chosen so that the calculation works (1 kW, 73 m/s)	Reality may be different	Medium	External review, explicit margins
Handling error during tests	Laser or high voltage without protocol	Personal injury	High	Written protocol, work in pairs, emergency stop

6.6 Risk analysis for the participants

Probability / severity, each rated out of 5 (5 = maximum).

Problem	Prob. / Severity	Solution
Loss or breakage of a computer	2/4	Backup device, handwriting
Long-term illness	3/2	Share the work, work more after recovery
Inability to work	2/5	Scale the project down, increase each person's workload
Exclusion from the master's programme	1/5	Same as above
Work overload (other courses, exams, internship)	3/4	Rely on a good schedule
Incompatible availabilities between members	4/3	Video meetings and/or state availabilities in advance, read the meeting minutes afterwards
Poor workload estimation	3/3	Redistribute the work among us, report it in a meeting or to the coordinator
Working on the same parts	4/4	Define the tasks with the WBS
Timing problem for deliverables	3/5	Respect the milestones set at the beginning of the year

6.7 Global table of the LiDAR + LASER payload

Origin	Main risk	Level	Main measure
Laser	Power and heat too high for a 6U	High	Reduce P_{avg}, thermal study
Laser	Poor pointing of the 3 mm spot at 50 m	High	Fine pointing mirror guided by the LiDAR
Laser	Debris drifts away during firing	High	Short shots, debris tracking, smaller ΔV
Laser	Debris fragmentation	High	Fluence control, automatic stop
Laser	Hazard to people during tests	High	Laser safety officer, goggles, closed room

Origin	Main risk	Level	Main measure
LiDAR	Loss of debris tracking	High	Reacquisition, adapted field of view
LiDAR	Weak signal or dazzled sensor	Medium	Link budget, 1550 nm filter
LiDAR	Millimetre accuracy not justified	Medium	Announce centimetre accuracy
Payload	Mass, volume and power incompatible with a 6U	High	Common budget with the other teams
Payload	LiDAR / laser misalignment	High	Rigid structure, in-orbit calibration
Payload	Firing software malfunction	High	Simple logic, tests
Payload	Launcher safety requirements and laser cost	High	Laser disabled before separation, supplier quotes
Human	Lack of laser and safety skills	High	Training, expert review
Human	Unnoticed calculation errors and poor communication between teams	Medium	Cross-review, interface meetings
Team	Working on the same parts (4/4)	High	Clear split with the WBS
Team	Delay on deliverables (3/5) and work overload (3/4)	High	Milestones set at the beginning of the year, realistic schedule
Team	Incompatible availabilities (4/3) and unavailability of a member	Medium	Video calls, minutes, work sharing

7 Project Management

7.1 Deliverables checklist

Deliverable	Section	Status
Definitive choice: studied object, precise part of the object, or functional mission target	1.1	**[TO DO:]**
Definitive choice: precise scientific / technological question or aim	1.2	**[TO DO:]**
Definitive choice: main payload	1.3	**[TO DO:]**
Definitive choice: secondary payloads (if any)	1.4	**[TO DO:]**
Ratio of contacted experts (X % out of the Y experts initially identified)	1.5	**[TO DO:]**
Precise functional description of all payloads' behavior	2.1	**[TO DO:]**
Precise quantified description of the payloads' performance	2.2	**[TO DO:]**
Data Management Plan functional diagram V1 (Miro.com)	3.1	**[TO DO:]**
Satellite platform functional diagram V1	3.2	**[TO DO:]**
Satellite system model V1 – IDM-CIC	3.3	**[TO DO:]**
Mission analysis V1 (orbit, phases)	4	**[TO DO:]**
Mission report summary V1	5	**[TO DO:]**
PPR tab elements 100 % done	–	**[TO DO:]**
Simple PS/OR slide deck ready (including screenshots of all up-to-date PM tools)	–	**[TO DO:]**